

LITERATURE REVIEW

A. Foundational Surveys

Title	Author(s)	Methodology	Gap related to your project
Domain Adaptation for the Classification of Remote Sensing Data: An Overview	Tuia, Persello & Bruzzone (2016)	Literature synthesis and taxonomy of DA techniques (invariant features, feature-space adaptation, active learning) for RS classification.	Predates GAN-based and self-supervised DA entirely — no coverage of contrastive learning, continuous embeddings, or generative cross-sensor translation.
The Eyes of the Gods: A Survey of UDA Methods Based on Remote Sensing Data	Xu, Wu, Chen, Zhang & Guo (2022)	Systematic review with a taxonomy of generative, adversarial, self-training, and hybrid UDA methods; defines partial, open-set, and multi-domain UDA.	Categorizes multi-domain UDA conceptually but proposes no method and no empirical unseen-pair generalization test — exactly the gap your project fills empirically.
Domain Adaptation for Satellite Images: Recent Advancements, Challenges, and Future Perspectives	Khazri Khelif, Boulila, Koubaa & Farah (2024)	Review focused specifically on self-supervised-learning-based DA for satellite imagery.	Survey-level only — no architecture, no test of generalization to unseen sensor-pair combinations.

B. Self-Supervised / Contrastive Multi-Sensor Representation Learning

Title	Author(s)	Methodology	Gap related to your project
Seasonal Contrast (SeCo)	Mañas, Lacoste, Giró-i-Nieto, Vazquez & Rodríguez (2021)	MoCo-style contrastive pretraining treating multi-temporal views of the same location as positive pairs, trained on uncured global Sentinel-2 data.	Single-sensor only — doesn't handle the SAR-vs-optical cross-sensor gap your encoder needs to learn.
Geography-Aware Self-Supervised Learning (GASSL)	Ayush, Uz Kent, Meng, Tanmay, Burke, Lobell	Contrastive learning using geolocation/season as metadata for positive pairs, plus a geo-location prediction auxiliary task.	Also single-sensor, no continuous cross-sensor embedding space, no generative component.

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	& Ermon (2021)		
SatMAE	Cong, Khanna, Meng, Liu, Rozi, He, Burke, Lobell & Ermon (2022)	ViT-based masked autoencoder with temporal/multi-spectral positional encodings, reconstructs masked patches.	Reconstruction-based, not contrastive; stays within one sensor family — no cross-sensor conditional generation.
Self-Supervised RS Scene Representations via Contrastive Multiview Coding	Stojnić & Risojević (2021)	Applies Contrastive Multiview Coding to RS scene classification, studies effect of pretraining domain on downstream accuracy.	Confirms RS-specific pretraining helps, but only for single-sensor scene classification — no multi-sensor or generative angle.

C. GAN-Based Cross-Sensor / Multi-Domain Translation

Title	Author(s)	Methodology	Gap related to your project
CycleGAN	Zhu, Park, Isola & Efros (2017)	Unpaired image translation via cycle-consistency + adversarial loss between exactly two domains.	Strictly pairwise — needs a new model per domain pair, no shared embedding space, no N-domain scaling. This is what your Phase 1 baseline exposes.
StarGAN	Choi, Choi, Kim, Ha, Kim & Choo (2018)	Single generator conditioned on a discrete one-hot target-domain label, using a domain classifier + adversarial loss.	Domain labels are fixed, discrete categories — not continuous embeddings learned without supervision.
StarGAN v2	Choi, Uh, Yoo & Ha (2020)	Learned style codes from a mapping/style network, AdaIN-conditioned generator for diverse multi-domain synthesis.	Style codes still come from labeled domain categories, not unlabeled satellite sensor data — your project's architectural base, extended to the RS domain gap.

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D. Dataset Papers

Title	Author(s)	Methodology	Gap related to your project
Panoptic Segmentation of Satellite Image Time Series (PASTIS, extended to PASTIS-HD)	Garnot & Landrieu (2021)	Convolutional temporal attention network for panoptic segmentation; establishes the labeled benchmark.	Dataset/benchmark paper, not a DA method — used purely as your primary labeled corpus (Objectives 2 & 4).
SpaceNet 6	Shermeyer, Hogan, Brown, Van Etten et al. (2020)	Dataset + baseline pairing SAR and optical imagery over Rotterdam with building-footprint labels.	Single-city — confirms the geographic-confound reasoning behind restricting it to zero-shot testing only.
MultiEarth 2022	Cha, Huang, Schmidt, Angelides, Hamilton et al. (2022)	Challenge paper defining three sub-tasks over a 4-sensor (S1, S2, Landsat-5, Landsat-8) Amazon time series with deforestation labels.	Dataset/challenge paper, no generator architecture proposed — used as your only 4-sensor source for the leave-one-pair-out test.

E. Multi-Domain Translation / Generalization to Unseen Combinations

Title	Author(s)	Methodology	Gap related to your project
Self-supervised Domain-agnostic Domain Adaptation for Satellite Images	Zhang, Shi & Zhu (2023)	Contrastive self-supervised GAN for domain-agnostic pairwise satellite image translation, no predefined domain labels.	My base paper — restricted to two-domain pairwise translation, no N-domain or unseen-pair mechanism.
StandardGAN (<i>published, IEEE CVPRW — corrected</i>)	Tasar, Tarabalka, Giros, Alliez & Clerc (2020)	GAN-based standardization of each source/target domain's distribution, then a single classifier trained on standardized data.	Normalizes distributions rather than learning a continuous embedding; handles one target at a time, not arbitrary many-to-many translation.

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Multi-Sensor Diffusion-Driven Optical Image Translation <i>(published, IEEE JSTARS — corrected)</i>	Vinholi, Chini, Amziane, Machado, Silva & Matgen (2024)	DDIM-based diffusion model super-resolving low-res imagery into high-res cross-sensor equivalents, tested on change detection.	Diffusion-based and computationally heavy; pairwise per-sensor training, no unseen sensor-pair generalization test.
Universal Multi-Domain Translation via Diffusion Routers <i>(now ICLR 2026 — was preprint)</i>	Kieu, Do, Hoang, Le, Kieu, Nguyen & Nguyen (2025)	Single diffusion noise predictor conditioned on domain labels, routes translation through a central domain using K-1 paired datasets.	General-purpose (natural images, not RS); still requires each domain paired with a central domain — confirms the unseen-combination problem is open outside RS too, supporting your novelty claim without solving it for satellite data.